

# CS 6341 Robotics: Instructions on Running ROS 2 with Docker on Windows, Mac and Ubuntu

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In this course, we will use **ROS 2 Jazzy Jalisco**. For Windows and Mac users, we recommend running ROS 2 inside Docker using the following Docker image:

<https://github.com/Tiryoh/docker-ros2-desktop-vnc>

This Docker image provides an Ubuntu desktop environment with ROS 2 installed. The desktop can be accessed directly from a web browser, so no additional X server software is required.

**Important:** Some PDF viewers may insert extra spaces when copying commands. For long Docker commands, we recommend copying the command from the course website rather than directly from this PDF. If you copy a command from the PDF, make sure that options such as `-shm-size`, `-name`, and addresses such as `ghcr.io` do not contain extra spaces.

## 1 For Windows and Mac Users: Using Docker

### 1.1 Step 1: Install Docker Desktop

- Download and install Docker Desktop: <https://www.docker.com/products/docker-desktop/>
- Start Docker Desktop.
- On Windows, make sure Docker Desktop is configured to use the WSL 2 backend if prompted.
- Open a terminal:
  - On Windows, use PowerShell or Windows Terminal.
  - On Mac, use Terminal.
- Verify that Docker is installed correctly:

```
docker --version
```

## 1.2 Step 2: Create a Data Folder on Your Host Computer

For this course, you should store your ROS 2 workspace and other course files on your **host computer**, rather than only inside the Docker container.

We will create a general data folder and mount the entire folder into Docker. Your ROS 2 workspace will be stored inside this folder. You can also use the data folder to store datasets, ROS bags, models, and other files used in the course.

This ensures that your files remain available even if the Docker container is accidentally deleted.

### Windows

Create a folder named:

```
C:\data
```

Then create a `ros2_ws` folder inside it:

```
C:\data\ros2_ws
```

You can create these folders using File Explorer or PowerShell.

### Mac

Open Terminal and run:

```
mkdir -p ~/data/ros2_ws
```

Your ROS 2 workspace will therefore be stored at:

```
~/data/ros2_ws
```

Other course files can also be stored under the data directory.

## 1.3 Step 3: Download and Run the ROS 2 Jazzy Docker Image

We will use the ROS 2 Jazzy image provided by:

<https://github.com/Tiryoh/docker-ros2-desktop-vnc>

When we create the container, we use the `-v` option to mount the data folder on your host computer into the Docker container.

### Windows PowerShell:

```
docker run -p 6080:80 --shm-size=512m --name ros2-jazzy -v C:\data:/home/ubuntu/data ghcr.io/tiryoh/ros2-desktop-vnc:jazzy
```

### Mac Terminal:

```
docker run -p 6080:80 --shm-size=512m --name ros2-jazzy -v ~/data:/home/ubuntu/data ghcr.io/tiryoh/ros2-desktop-vnc:jazzy
```

**Important:** We recommend copying these long commands from the course website instead of directly from the PDF.

The first time you run the command, Docker will automatically download the ROS 2 Jazzy image. This may take some time depending on your Internet connection.

The option

```
--name ros2-jazzy
```

assigns the name `ros2-jazzy` to the container.

The option

```
-v HOST_DATA_PATH:/home/ubuntu/data
```

mounts the data directory on your host computer to

```
/home/ubuntu/data
```

inside the Docker container.

Therefore, for example:

```
C:\data\ros2_ws
```

on Windows, or

```
~/data/ros2_ws
```

on Mac corresponds to:

```
/home/ubuntu/data/ros2_ws
```

inside Docker.

## 1.4 Step 4: Open the Ubuntu Desktop

After the Docker container starts, open a web browser and go to:

```
http://127.0.0.1:6080/
```

You should see an Ubuntu desktop environment running inside your browser.

ROS 2 Jazzy is already installed in this environment. You do **not** need to install ROS 2 manually.

## 1.5 Step 5: Open a Terminal Inside Ubuntu

Open a terminal from the Ubuntu desktop.

Before using ROS 2, run:

```
source /opt/ros/jazzy/setup.bash
```

Verify that ROS 2 Jazzy is active:

```
echo $ROS_DISTRO
```

You should see:

```
jazzy
```

You can also verify ROS 2 by running:

```
ros2 --help
```

## 1.6 Step 6: Test the ROS 2 Installation with RViz

In the Ubuntu terminal, run:

```
source /opt/ros/jazzy/setup.bash  
rviz2
```

The RViz window should appear inside the Ubuntu desktop in your browser.

## 1.7 Step 7: Set Up Your ROS 2 Workspace

Your host data directory is available inside Docker at:

```
/home/ubuntu/data
```

Your ROS 2 workspace is therefore:

```
/home/ubuntu/data/ros2_ws
```

Create the src directory:

```
mkdir -p ~/data/ros2_ws/src
```

Your workspace should now have the following structure:

```
~/data/ros2_ws/  
  src/
```

ROS 2 packages used in the course should be placed inside:

```
~/data/ros2_ws/src
```

Because the data directory is mounted from your host computer, files that you create or modify here are also stored on your Windows or Mac computer.

## 1.8 Step 8: Add a ROS 2 Package to the Workspace

We will use the ROS 2 `urdf_tutorial` package to test the workspace.

The package is available at:

`https://github.com/ros/urdf\_tutorial`

Go to the `src` directory:

```
cd ~/data/ros2_ws/src
```

Clone the ROS 2 branch of the package:

```
git clone -b ros2 https://github.com/ros/urdf_tutorial.git
```

You may also need to install the `urdf_launch` package:

```
sudo apt update
sudo apt install ros-jazzy-urdf-launch
```

After cloning, your workspace should look similar to:

```
~/data/ros2_ws/
src/
  urdf_tutorial/
```

## 1.9 Step 9: Build the ROS 2 Workspace

Before building the workspace, source ROS 2:

```
source /opt/ros/jazzy/setup.bash
```

Go to the root of the workspace:

```
cd ~/data/ros2_ws
```

Then build:

```
colcon build
```

**Important: Do not use `sudo colcon build`.**

After a successful build, you should see the following directories:

```
~/data/ros2_ws/
build/
install/
log/
src/
```

Source the newly built workspace:

```
source ~/data/ros2_ws/install/setup.bash
```

Verify that the urdf\_tutorial package can be found:

```
ros2 pkg prefix urdf_tutorial
```

The command should return the installation location of the package.

Test the urdf tutorial package

```
ros2 launch urdf_tutorial display.launch.py
```

## 1.10 Step 10: Automatically Source ROS 2

Instead of running

```
source /opt/ros/jazzy/setup.bash
```

every time you open a new terminal, you can add it to your .bashrc file:

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

After you have successfully built your ROS 2 workspace, you may also add:

```
echo "source ~/data/ros2_ws/install/setup.bash" >> ~/.bashrc
```

## 1.11 Step 11: Stop the Docker Container

If Docker is running in the foreground in your Windows or Mac terminal, press:

Ctrl+C

You can also stop the container from another terminal using:

```
docker stop ros2-jazzy
```

## 1.12 Step 12: Start the Container Again

You only need to use `docker run` when creating the container for the first time.

After the container has been created, start the same container using:

```
docker start ros2-jazzy
```

Then open:

<http://127.0.0.1:6080/>

You do **not** need to specify the `-v` option again. The volume mount that you specified when creating the container will automatically be used again.

To see all Docker containers, run:

```
docker ps -a
```

To see currently running containers, run:

```
docker ps
```

## 1.13 Step 13: Understand What Is Stored Where

The entire data directory is mounted from your host computer into Docker:

Windows/Mac data  $\longleftrightarrow$  Docker /home/ubuntu/data

For example, your ROS 2 workspace is stored as:

C:\data\ros2\_ws (Windows)  $\longleftrightarrow$  /home/ubuntu/data/ros2\_ws

or:

~/data/ros2\_ws (Mac)  $\longleftrightarrow$  /home/ubuntu/data/ros2\_ws

You can create additional directories under data for datasets, ROS bags, models, or other course files. These files will also be stored on your host computer.

Packages and libraries installed using commands such as

```
sudo apt install <package>
```

or

```
pip install <package>
```

are normally installed **inside the Docker container**. They remain available when you stop and restart the same container:

```
docker stop ros2-jazzy
```

```
docker start ros2-jazzy
```

However, packages installed inside the container will be lost if you delete the container using:

```
docker rm ros2-jazzy
```

Therefore:

- Keep your ROS 2 workspace, datasets, ROS bags, models, and other important course files in the mounted data directory.
- Use the same ros2-jazzy container throughout the semester.
- Use `docker stop` and `docker start` to stop and restart the container.
- Do not delete the container unless you intend to recreate the software environment.

## 2 For Ubuntu Users: Native ROS 2 Installation

If you are using **Ubuntu 24.04**, we recommend installing ROS 2 Jazzy directly on your computer instead of using Docker.

Follow the official ROS 2 Jazzy installation instructions:

<https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html>

We recommend installing the ROS 2 Desktop version and development tools:

```
sudo apt install ros-jazzy-desktop
sudo apt install ros-dev-tools
```

After installation, source ROS 2:

```
source /opt/ros/jazzy/setup.bash
```

Verify the installation:

```
echo $ROS_DISTRO
```

The output should be:

```
jazzy
```

You can also test the installation with RViz:

```
rviz2
```

After ROS 2 Jazzy is installed successfully, follow **Steps 7–10 in the Windows and Mac section** to create the ROS 2 workspace, add the `urdf_tutorial` package, build the workspace, and configure your ROS 2 environment.

For Ubuntu users, `~/data` is a normal directory on your computer and does not need to be mounted using Docker.

## 3 Troubleshooting

### 3.1 Docker Command Does Not Work After Copying from the PDF

Some PDF viewers may insert spaces into commands when copying text.

For example, the following is **incorrect**:

```
--shm - size =512 m
-- name ros2 - jazzy
ghcr . io / tiryoh / ros2 - desktop - vnc : jazzy
```

The correct text should contain no extra spaces:

```
--shm-size=512m
--name ros2-jazzy
ghcr.io/tiryoh/ros2-desktop-vnc:jazzy
```

For long Docker commands, copy the command from the course website whenever possible.

### 3.2 A Container Named `ros2-jazzy` Already Exists

If you try to run `docker run` again, Docker may report that the container name `ros2-jazzy` is already in use.

Normally, you should simply start your existing container:

```
docker start ros2-jazzy
```

Do not create a new container every time you use ROS 2.

### 3.3 RViz or Other GUI Applications Do Not Display Correctly

The Docker image normally supports graphical ROS applications directly through the browser.

If RViz, Gazebo, or another GUI application shows a blank window or does not start correctly, first make sure that Docker Desktop is up to date.

If the problem persists, the Docker image documentation suggests using `seccomp=unconfined`. Note that this reduces Docker's default container isolation.

If you recreate the container, remember to include your `-v data` directory mount in the new `docker run` command.

### 3.4 ROS 2 Packages Cannot Find `ament_cmake`

Make sure ROS 2 has been sourced:

```
source /opt/ros/jazzy/setup.bash
```

Then build your workspace without `sudo`:

```
cd ~/data/ros2_ws
colcon build
```

Do **not** run:

```
sudo colcon build
```

Using `sudo` can remove ROS-related environment variables and cause CMake to fail to find packages such as `ament_cmake`.

### 3.5 Check the ROS 2 Version

Run:

```
echo $ROS_DISTRO
```

The output should be:

```
jazzy
```

## 4 Summary

For Windows and Mac users, the recommended workflow for this course is:

1. Install Docker Desktop.
2. Create a data directory on your host computer.
3. Create `ros2_ws` inside the data directory.
4. Create the ROS 2 Jazzy Docker container and mount the entire data directory using `-v`.
5. Open `http://127.0.0.1:6080/` in a browser.
6. Use the Ubuntu desktop and terminal inside the browser.
7. Create the ROS 2 workspace structure.
8. Clone the `urdf_tutorial` package into `src`.
9. Build the workspace using `colcon build` without `sudo`.
10. Source the workspace and verify the package.
11. Stop the container using `docker stop ros2-jazzy`.
12. Restart it later using `docker start ros2-jazzy`.

This setup provides the same ROS 2 Jazzy environment for both Windows and Mac users while keeping course files safely stored on the host computer.